

❖ Section 2. Discrete Markov Chains

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- Discrete time, discrete state space
- Markov property
- Transition matrix P
- Transition probability

History of Markov Chain:

Andrei Markov (1856 - 1922) founded the modern theory of stochastic processes, with this special case named as Markov chains.

Many real systems evolve randomly **over time**, but:

The future depends **only on the present**, not the full past.

Markov Chains $\leftrightarrow$ Probability + Linear Algebra + Graph Theory

The theory has an enormous range of applications including: statistical physics, queueing theory, communication networks, voice recognition, bioinformatics, Google's page-rank algorithm, machine learning and inference, economics, gambling, data compression.

➤ Discrete-time Markov Chain

Let Ω be a finite or countably infinite sample space.

A **Discrete-time Markov chain** is a stochastic model describing a sequence of possible events in which the probability of future event X_{t+1} depends only on the **state** attained in the previous event X_t , not on how it got to its present position.

$$X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots \rightarrow X_{t-1} \rightarrow X_t \rightarrow X_{t+1}, \dots$$

Examples: X_t =weather on day t . State space: $\Omega = \{sunny, rainy, cloudy\}$

X_t =position of a particle after t steps. $\Omega = \{(x, y) \in \mathbb{R} \times \mathbb{R}\}$

X_t =gambler's capital after t rounds. $\Omega = \{0, 1, \dots, N\}$

The Markov condition

Let $\Omega = \{1, 2, \dots, i, \dots, j, \dots\}$ be a finite or countably infinite **sample space**.

Definition: A collection of Ω -valued random variables

$$\{X_0, X_1, X_2, \dots, X_{t-1}, X_t, X_{t+1}, \dots\}$$

is called a **discrete-time Markov Chain** on Ω if it satisfies the Markov condition:

$$P(X_{t+1} = j | X_t = i, X_{t-1} = i_{t-1}, \dots, X_0 = i_0) = P(X_{t+1} = j | X_t = i)$$

for all $n \geq 0$ and all **states** $j, i_0, \dots, i_n \in \Omega$.

- Markov conditions says that, Given X_t , the future X_{t+1} is independent of the past history $X_0, \dots, X_{t-1}$.
- If $X_t = i$, we say that the chain visits state i , or hits i at time t .

Example:

X_n = Stock price

Ω = range of possible values of stock price

$= \{0, 0.01, 0.02, \dots, 0.11, \dots, 1.01, \dots 2.01, \dots\}$

Example:

X_n = Word in sentence (Speech recognition)

Ω = Set of possible words = {word1, word2, ...}

Remark: Markov condition is a strong structural assumption. It is never exactly true in reality.

Remarks:

Markov chain:

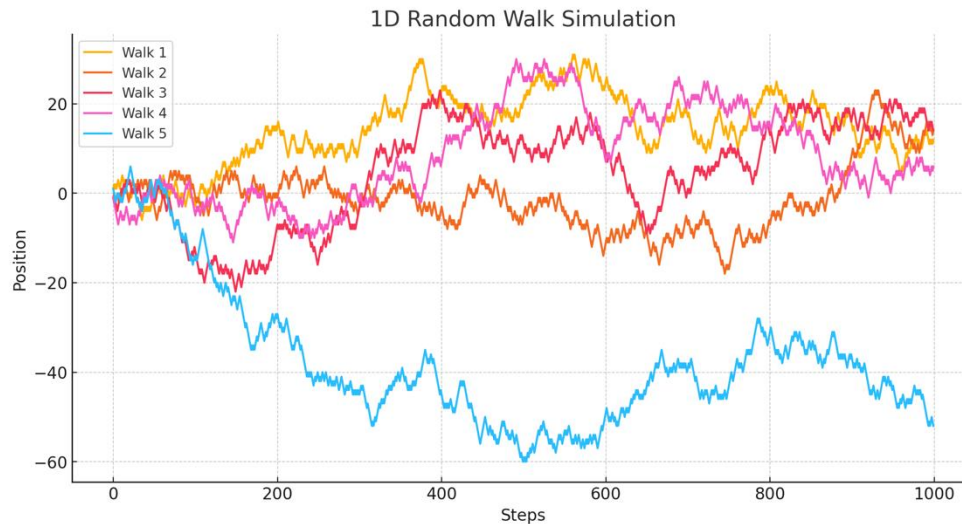
$$X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots \rightarrow X_{n-1} \rightarrow X_n \rightarrow X_{n+1}, \dots$$

Sequence of Ω -valued random variables, discrete time indexed by integer subscripts, related by conditional probabilities between successive variables.

- The sequence $\{X_0, X_1, X_2, \dots\}$ is dependent but in a simplest possible way.
- The future (random) behavior of the chain only depends on where the chain sits right now, and not on how it got to its present position.
- They are the nearest to independent chains.

Example. The sequence of **random steps** $Y_1, Y_2, \dots$ independent, while the **random walk** $X_k = Y_1 + \dots + Y_k$ is not independent. However, they are both Markov chains.

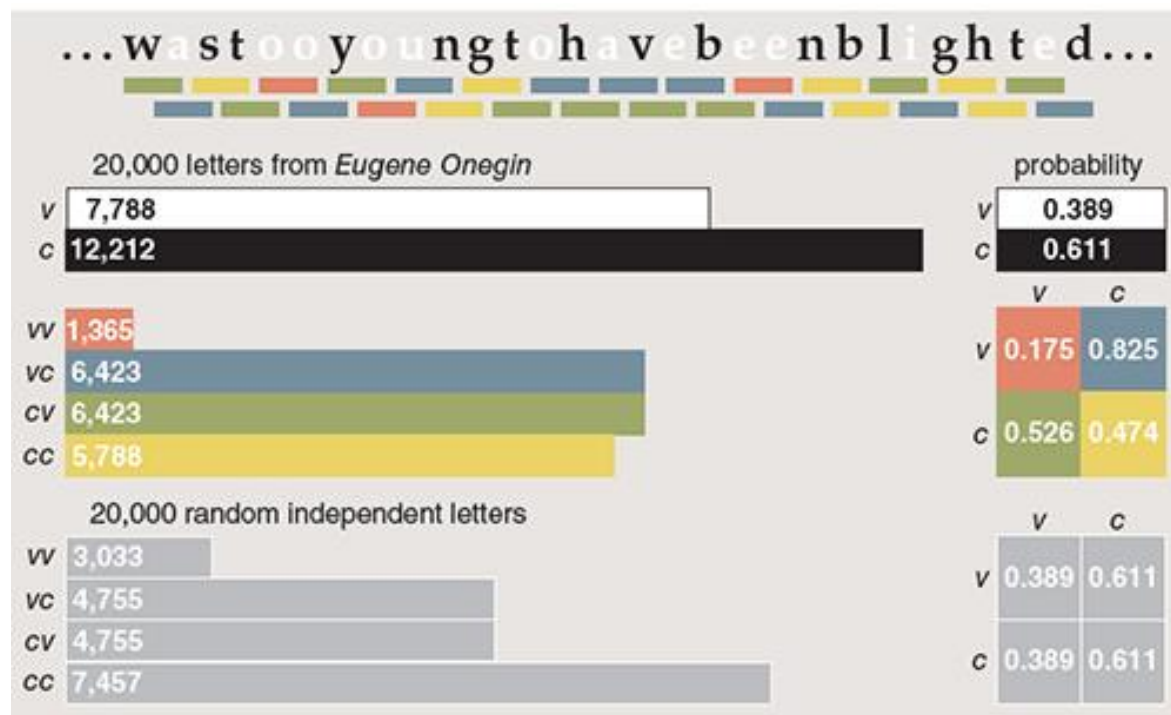
$$X_{k+1} = X_k + Y_{k+1}$$



A famous example: (Counting Vowels and Consonants)

A. A. Markov studied the sequence of 20,000 letters in A. S. Pushkins poem “Eugeny Onegin”, discovering that the **stationary vowel** probability is $p = 0.389$, that the probability of a **vowel** following a **vowel** is $p_1 = 0.175$, and that the probability of a **vowel** following a **consonant** is $p_2 = 0.526$.

*He was too young to have been blighted
by the cold world's corrupt finesse;
his soul still blossomed out, and lighted
at a friend's word, a girl's caress.
In heart's affairs, a sweet beginner,
he fed on hope's deceptive dinner;
the world's éclat, its thunder-roll,
still captivated his young soul.
He sweetened up with fancy's icing
the uncertainties within his heart;
for him, the objective on life's chart
was still mysterious and enticing—
something to rack his brains about,
suspecting wonders would come out.*



□ Transition Matrix:

- The **matrix** P describing the Markov Chain with entries p_{ij} , and

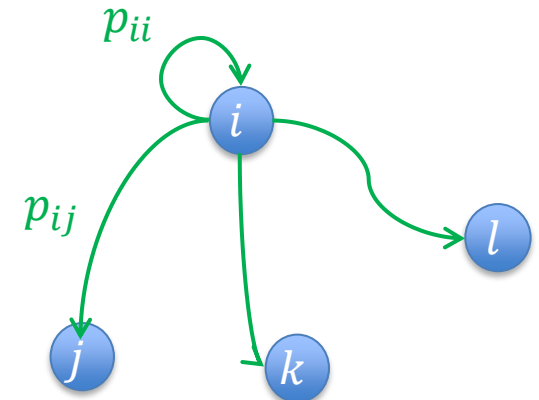
$$p_{ij} := \mathbb{P}(X_{n+1} = j \mid X_n = i) \quad \text{for all } n \text{ and } i, j \in \Omega$$

Time-homogeneous assumption: p_{ij} does not depend on n .

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i) = \cdots = \mathbb{P}(X_1 = j \mid X_0 = i)$$

is called the one-step **transition probability**,

$$P = \begin{bmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mm} \end{bmatrix}$$



Any **transition matrix** P must satisfy these properties:

- $p_{ij} \geq 0$ for all states $i, j \in \Omega$
- $\sum_{j \in \Omega} p_{ij} = 1$ for all $i \in \Omega$ (Each i -th **row** of P should each sum to 1)

Such a matrix is also called **row**-stochastic. So, a square matrix is a transition matrix if and only if it is non-negative and **row**-stochastic.

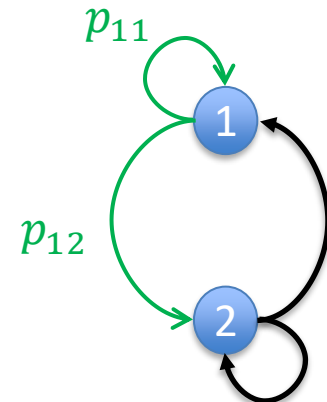
Example:

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$$

$$p_{11} := P(X_{n+1} = 1 | X_n = 1)$$

$$p_{12} := P(X_{n+1} = 2 | X_n = 1)$$

$$p_{11} + p_{12} = 1$$



Verify the transition matrix property:

For all $i \in \Omega$,

$$\begin{aligned}\sum_{j \in \Omega} p_{ij} &= \sum_{j \in \Omega} \mathbb{P}(X_1 = j \mid X_0 = i) \\&= \sum_{j \in \Omega} \frac{\mathbb{P}(X_1 = j, X_0 = i)}{\mathbb{P}(X_0 = i)} \\&= \frac{\sum_{j \in \Omega} \mathbb{P}(X_1 = j, X_0 = i)}{\mathbb{P}(X_0 = i)} \\&= \frac{\mathbb{P}(X_0 = i)}{\mathbb{P}(X_0 = i)} \\&= 1\end{aligned}$$

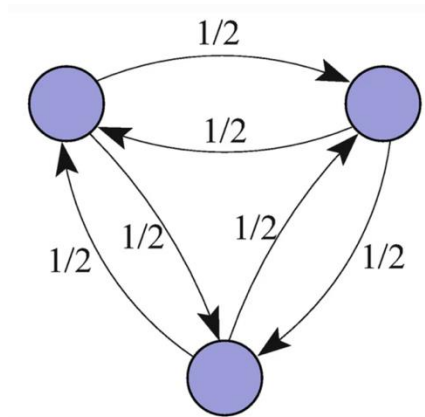
Example (Counting Vowels and Consonants).

In the previous Counting Vowels and Consonants example, the transition matrix is

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} 0.175 & 0.825 \\ 0.526 & 0.474 \end{bmatrix}$$

The transition probability $p_{12} = P(X_{n+1} = 2 \text{ consonant} | X_n = 1 \text{ vowel})$

Examples:



$$T = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

Question: What is the transition matrix for an IID sequence?

$$X_0, X_1, X_2, \dots, X_t, \dots$$

Assume state space is $\Omega = \{1, 2, \dots, m\}$

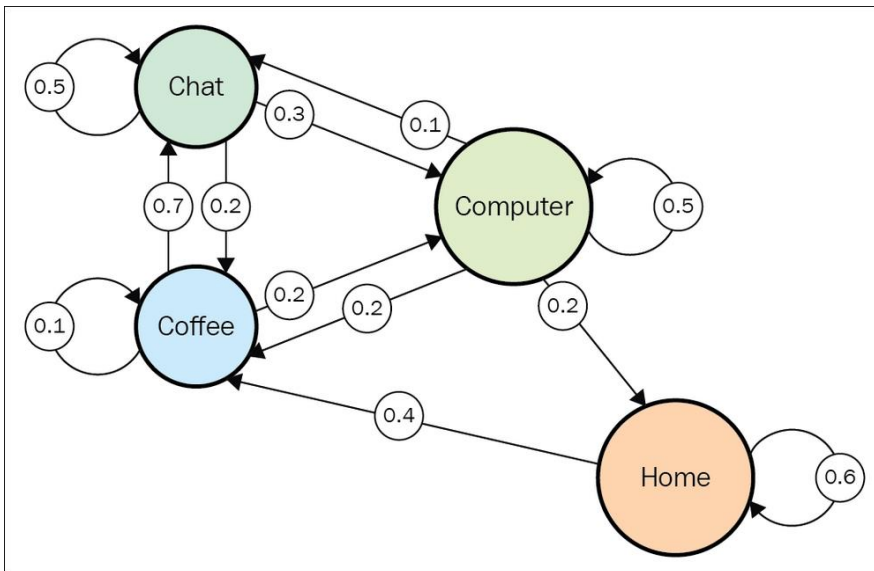
Assume $\mathbb{P}(X_n = i) = p_i$ for $i = 1, \dots, m$ and $n \geq 0$.

The total probability $p_1 + \dots + p_m = 1$

Example: Office Worker

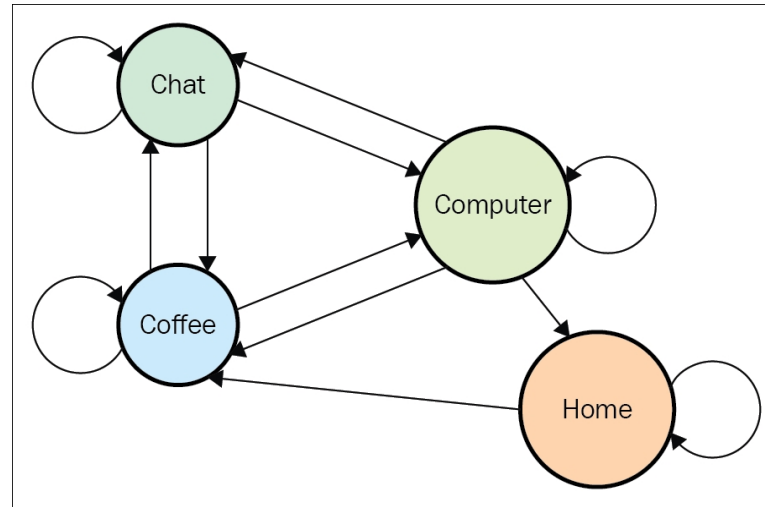
State Space: $\Omega = \{1,2,3,4\}$

1. **Home:** He's not at the office
2. **Computer:** He's working on his computer at the office
3. **Coffee:** He's drinking coffee at the office
4. **Chatting:** He's discussing something with colleagues at the office



	Home	Coffee	Chat	Computer
Home	60%	40%	0%	0%
Coffee	0%	10%	70%	20%
Chat	0%	20%	50%	30%
Computer	20%	20%	10%	50%

In practice, we rarely have the luxury of knowing the exact transition matrix.



A much more real-world situation is when we have only observations of our systems' states, which are also called episodes:

- home → coffee → coffee → chat → chat → coffee → computer → computer → chat → computer → computer → chat → chat → coffee → computer → computer → chat → coffee → chat → computer → coffee → coffee → computer → computer → home

It's not complicated to estimate the transition matrix by our observation; we just **count** all the transitions from every state and normalize them to a sum of 1. The more observation data we have, the closer our estimation will be to the true underlying model.

Question: Is a process with memory length 2 Markov?

Example: Suppose “Weather (Sunny, Rainy)” process has **2-day memory**

$$\mathbb{P}(\text{rain tomorrow} \mid \text{history}) = \mathbb{P}(\text{rain tomorrow} \mid \text{today, yesterday})$$

- If it has rained for the past two days, then it will rain tomorrow with probability 0.7;
- If it rained today but not yesterday, then it will rain tomorrow with probability 0.5;
- if it rained yesterday but not today, then it will rain tomorrow with probability 0.4;
- if it has not rained in the past two days, then it will rain tomorrow with probability 0.2.

Then, **State**=(Sunny, Rainy) is not Markov.

$$P(X_{t+1} = R \mid X_t = R) \text{ is not well-defined.}$$

Question: Can we make it Markov by redefining the state?

Define the Markov Chain State as the **pair (today, yesterday)**.

- **state 0:** (today R, yesterday R) = RR
- **state 1:** (today R, yesterday N) = NR
- **state 2:** (today N, yesterday R) = RN
- **state 3:** (today N, yesterday N) = NN

So the transition matrix is

$$P = \begin{pmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{pmatrix}$$

Example. Gambler's Ruin. Random Walk with $X_0 = i$

X_t = gambler's capital after t rounds.

State space: $\{0, 1, 2, \dots, N\}$

Time: discrete $t = 0, 1, 2, \dots$

$$X_{t+1} = \begin{cases} X_t + 1 & \text{with probability } p \\ X_t - 1 & \text{with probability } q \end{cases}$$

Markov Property

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots) = P(X_{t+1} \mid X_t)$$

0

1

2

3

4

...

N

Absorbing states: 0 and N

$$p_{00} = 1, \text{ and } p_{NN} = 1$$

For interior states $i = 1, 2, \dots, N - 1$, (transient state)

$$p_{i,i+1} = p \text{ and } p_{i,i-1} = 1 - p$$

Transition matrix P is $(N + 1) \times (N + 1)$ with

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 1-p & 0 & p & 0 & \cdots & 0 & 0 \\ 0 & 1-p & 0 & p & \cdots & 0 & 0 \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ 0 & 0 & \cdots & 1-p & 0 & p & 0 \\ 0 & 0 & \cdots & 0 & 1-p & 0 & p \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix}.$$

❑ Probability of Success

Define $h(i) := P(\text{reach } N \text{ before } 0 \mid X_0 = i)$.

Boundary conditions

$$h(0) = 0, \text{ and } h(N) = 1.$$

Recurrence relation by conditioning on the first step:

$$h(i) = ph(i+1) + qh(i-1), \quad 1 \leq i \leq N-1,$$

- A **fair** game ($p = q = \frac{1}{2}$), the winning probability is proportional to initial capital

$$h(i) = \frac{i}{N}$$

Fair game $\neq$ safe game

- **Biased** game ($p \neq q$)

$$h(i) = \frac{1 - \left(\frac{q}{p}\right)^i}{1 - \left(\frac{q}{p}\right)^N}$$

- If $p < q$: ruin is almost certain for large N .
- If $p > q$: gambler has a genuine advantage.

Expected duration of the game

Let $m_i = E[\text{time to absorption} \mid X_0 = i]$.

This satisfies:

$$m_i = 1 + pm_{i+1} + (1 - p)m_{i-1},$$

$$m_0 = m_N = 0.$$

Special case: $p = \frac{1}{2}$

$$m_i = i(N - i)$$

Maximum duration when starting near $i = \frac{N}{2}$.

□ Transition Matrix as a Linear Operator

Matrix representation

$$P = \begin{bmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mm} \end{bmatrix} \quad \begin{aligned} p_{ij} &= \mathbb{P}(X_{n+1} = j | X_n = i) \\ &= \mathbb{P}(X_1 = j | X_0 = i) \end{aligned}$$

Suppose the initial distribution of X_0 is

$$\mu^{(0)} := \alpha := (\mathbb{P}(X_0 = 1), \dots, \mathbb{P}(X_0 = m))$$

Then, the distribution of X_1

$$\mu^{(1)} := (\mathbb{P}(X_1 = 1), \dots, \mathbb{P}(X_1 = m))$$

is

$$\mu^{(n)} = \mu^{(0)} P = \alpha P$$

Reason:

$$P(X_1 = 1) = \sum_{i=1}^m P(X_0 = i) P(X_1 = 1 | X_0 = i) = \sum_{i=1}^m \alpha_i p_{i1}$$

Distribution of X_n

More generally, the distribution evolution (row) vector of X_n

$$\mu^{(n)} := (\mathbb{P}(X_n = 1), \dots, \mathbb{P}(X_n = m))$$

Then, the distribution of X_{n+1} is

$$\mu^{(n)} = \mu^{(n-1)} P$$

Reason:
$$\begin{aligned} \mathbb{P}(X_n = j) &= \sum_{i=1}^m \mathbb{P}(X_{n-1} = i) \mathbb{P}(X_n = j | X_{n-1} = i) \\ &= \sum_{i=1}^m \mathbb{P}(X_{n-1} = i) \mathbb{P}(X_1 = j | X_0 = i) \end{aligned}$$

By induction, the distribution of X_{n+1} :

$$\mu^{(n)} = \mu^{(n-1)} P = \dots = \mu^{(0)} P^n = \alpha P^n$$

Special case when

$$e_i = (0, \dots, 0, 1, 0, \dots, 0)$$

$$e_i P =$$

$$e_i P^n =$$

Row vector Convention:

- Rows = current state
- Columns = next state

$$p_{ij} = P(X_{n+1} = j | X_n = i)$$

From state i to state j

In linear algebra textbook, people usually use **column** vector convention.

$$\vec{\mu}^{(n+1)} = P^T \vec{\mu}^{(n)}$$

$$\vec{\mu}^{(n)} = \begin{bmatrix} \mathbb{P}(X_n = 1) \\ \mathbb{P}(X_n = 2) \\ \dots \\ \mathbb{P}(X_n = m) \end{bmatrix}$$

As linear map of distributions: $\mathbb{R}^m \rightarrow \mathbb{R}^m$

$$\vec{\mu}^{(n)} \rightarrow \vec{\mu}^{(n+1)}$$

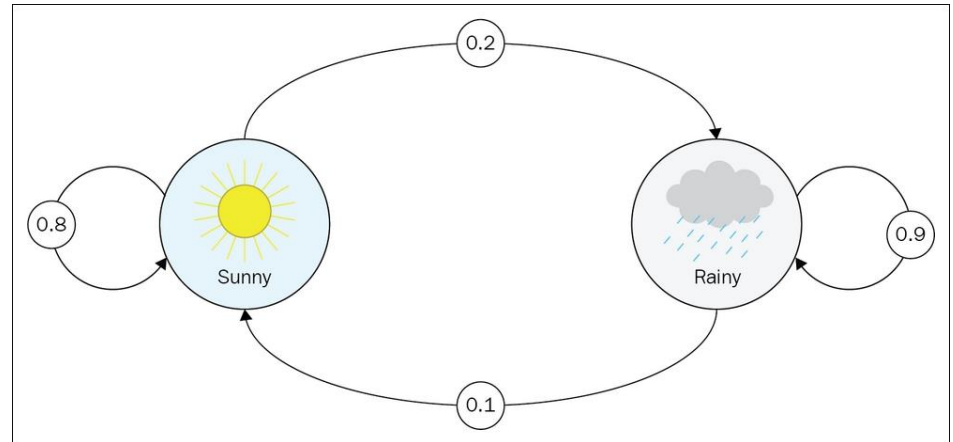
There is no mathematical difference — only a convention.

Example: Weather prediction:

X_t = weather on day t .

$\Omega = \{\text{sunny}, \text{rainy}\}$

	sunny	rainy
sunny	0.8	0.2
rainy	0.1	0.9



1. What does p_{21} mean?

2. If today is rainy, what is tomorrow's distribution?

3. If today's weather is uncertain, how do we compute tomorrow's?

How about the day after tomorrow?

Let $\mu^{(0)} = (0.6, 0.4)$

Then, $\mu^{(1)} = (0.6, 0.4)P =$

A Sample path (one year data in a Boston)

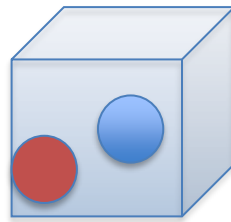
Fix $\omega: S \rightarrow R \rightarrow R \rightarrow S \rightarrow S \rightarrow R \rightarrow \dots \rightarrow S \rightarrow R \rightarrow R$

Distribution

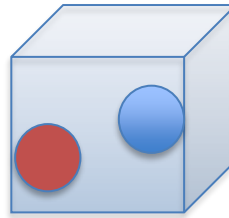
Across all $\omega: P(X_n = i)$

Example (Balls and Boxes Markov Chain).

There are four balls, two red and two blue, and two boxes. Initially, two balls are placed in each box. At each discrete time step, one ball is selected uniformly at random from each box; the two selected balls are exchanged and then placed back into the boxes.



Box1



Box2

Let X_n denote the number of red balls in the first box after n steps.

(a) Show that $\{X_n\}_{n \geq 0}$ is a Markov chain and identify its state space.

(b) Compute the one-step transition probabilities and write down the transition matrix P .

(a) Markov property

The state space is $\Omega = \{0,1,2\}$ since Box 1 contains 2 balls total, so it can have 0, 1, or 2 red balls.

Given $X_n = i$, the composition of the boxes is determined:

- Box 1 has i red and $2 - i$ blue.
- Box 2 has $2 - i$ red and i blue (because total red = 2).

The next step depends only on which colors are drawn from each box, and the probabilities of drawing a red/blue from a box depend only on i . Therefore,

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i)$$

so $\{X_n\}$ is a Markov chain.

0

1

2

(b) Transition probabilities and matrix

- From $i = 0$: Box 1 is BB , Box 2 is WW . You must swap a B with a R , so $X_{n+1} = 1$ with probability 1.

Hence $p_{01} = 1$, others 0.

- From $i = 2$: symmetric, so again $X_{n+1} = 1$ with probability 1.

Hence $p_{21} = 1$, others 0.

- From $i = 1$: each box is RB .

- To go to 0: need to swap R from Box 1 with B from Box 2. Probability $(1/2)(1/2) = 1/4$.
- To stay at 1: either swap R with R or B with B . Probability $1/4 + 1/4 = 1/2$.
- To go to 2: need to swap B from Box 1 with R from Box 2. Probability $(1/2)(1/2) = 1/4$.

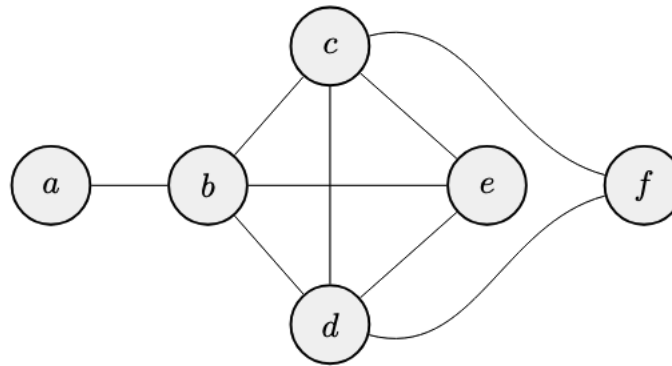
So the transition matrix (rows = current state i , columns = next state j) is

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix}$$

Example: Simple Random walk on a graph

A **graph** is a set of vertices and a set of edges. Two vertices are neighbors if there is an edge joining them. The **degree** of a vertex v , denoted $\deg(v)$, is the number of neighbors of v .

Example:



$$\deg(a) = 1, \deg(b) = \deg(c) = \deg(d) = 4, \deg(e) = 3, \deg(f) = 2.$$

At each discrete unit of time, a frog jumps to a neighboring pad **uniformly** at random.

Let X_n be the frog's location after n jumps. The sequence $X_0, X_1, \dots$ is a Markov chain. Given a graph G such a process is called **simple random walk** on G .

Write $i \sim j$ if vertices i and j are neighbors. The one-step transition probabilities:

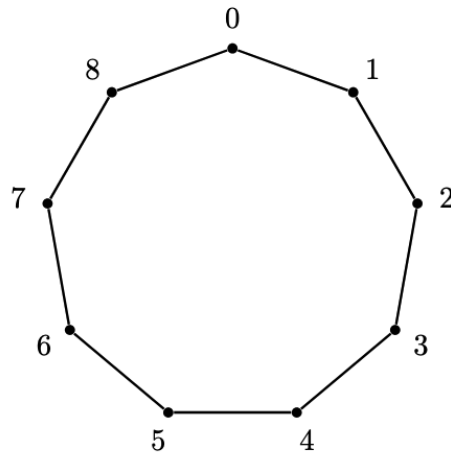
$$P(X_1 = j \mid X_0 = i) = \begin{cases} \frac{1}{\deg(i)}, & \text{if } i \sim j, \\ 0, & \text{otherwise.} \end{cases}$$

The transition matrix for simple random walk on the graph

$$P = \begin{array}{c|cccccc} & a & b & c & d & e & f \\ \hline a & 0 & 1 & 0 & 0 & 0 & 0 \\ b & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 \\ c & 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ d & 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ e & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ f & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \end{array}$$

More examples

(a) Cycle graph on nine vertices. Simple random walk on the cycle moves left or right with probability $\frac{1}{2}$. Each vertex has degree two.

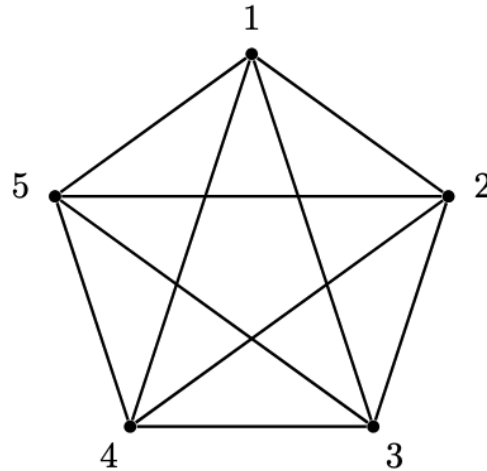


For a cycle with k vertices, the transition matrix is

$$P_{ij} = \begin{cases} \frac{1}{2}, & \text{if } j \equiv i \pm 1 \pmod{k}, \\ 0, & \text{otherwise.} \end{cases}$$

(b) Complete graph on five vertices

In the complete graph, every pair of vertices is joined by an edge. The complete graph on five vertices is shown.



The complete graph on k vertices has $\binom{k}{2}$ edges.

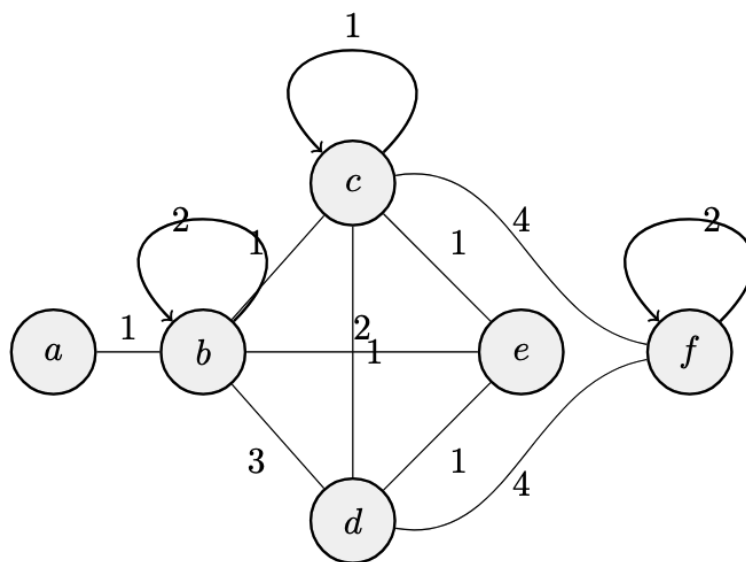
Each vertex has degree $k - 1$. The entries of the transition matrix are

$$P_{ij} = \begin{cases} \frac{1}{k-1}, & \text{if } i \neq j, \\ 0, & \text{if } i = j. \end{cases}$$

Example: Weighted, directed, graph

A **weighted graph** associates a positive number (**weight**) with every edge.

Example: The graph contains **loops** at vertices b, c , and f , which are edges joining a vertex to itself.



For random walk on a weighted graph, transition probabilities are proportional to the weights.

If vertices i and j are neighbors, let $w(i, j)$ denote the weight of the edge joining i and j .

Let the **weight degree** of i

$$w(i) = \sum_{k: i \sim k} w(i, k)$$

be the sum of the weights on all edges joining i to its neighbors.

The **transition probabilities** are

$$P_{ij} = \begin{cases} \frac{w(i, j)}{w(i)}, & \text{if } i \sim j, \\ 0, & \text{otherwise.} \end{cases}$$

The **transition matrix** of the example is

$$P = \begin{array}{c|cccccc} & a & b & c & d & e & f \\ \hline a & 0 & 1 & 0 & 0 & 0 & 0 \\ b & \frac{1}{9} & \frac{2}{9} & \frac{1}{9} & \frac{3}{9} & \frac{2}{9} & 0 \\ c & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{4}{8} \\ d & 0 & \frac{3}{9} & \frac{1}{9} & 0 & \frac{1}{9} & \frac{4}{9} \\ e & 0 & \frac{2}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ f & 0 & 0 & \frac{4}{10} & \frac{4}{10} & 0 & \frac{2}{10} \end{array}$$

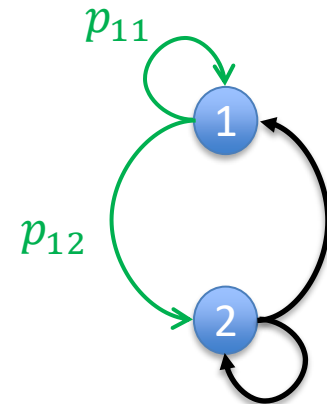
Transition matrix $\rightarrow$ weighted directed graph

Given a Markov transition matrix P , with state space $\Omega = \{1, 2, \dots, n\}$

Define a weighted directed graph $G = (V, E, w)$

- **Vertices:** $V = \{1, 2, \dots, n\}$
- **Directed edges:** Draw an arrow $i \rightarrow j$ **iff** $p_{ij} > 0$.
- **Edge weights:** Assign $w(i \rightarrow j) = p_{ij}$.

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$$

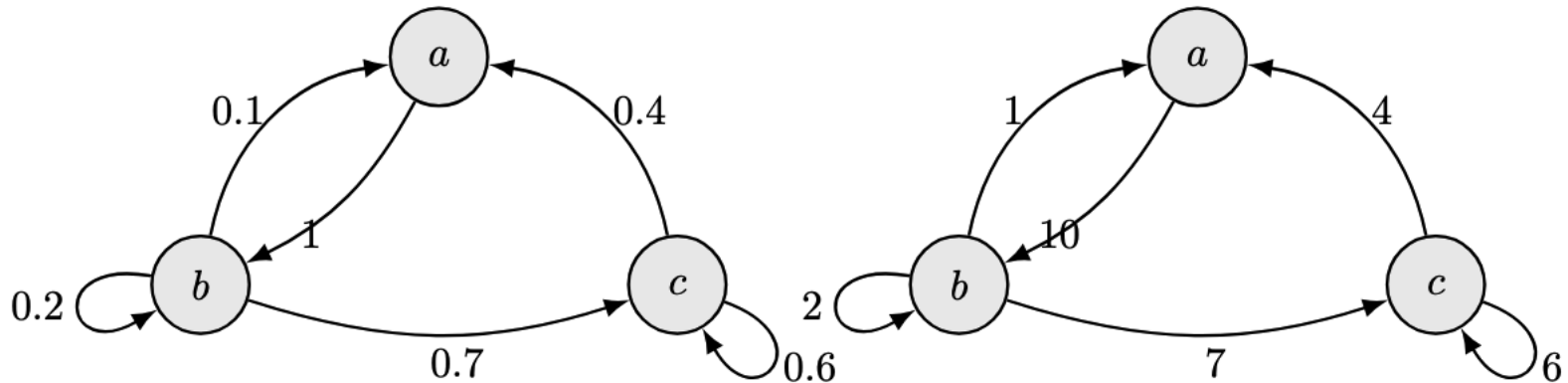


Example: From transition matrix to graph

Consider a transition matrix

$$P = \begin{array}{c|ccc} & a & b & c \\ \hline a & 0 & 1 & 0 \\ b & 0.1 & 0.2 & 0.7 \\ c & 0.4 & 0 & 0.6 \end{array} .$$

Two transition graphs corresponding to the same transition matrix.



❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross